

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
from google.colab import files

uploaded = files.upload()

# Get the actual filename from the 'uploaded' dictionary key
file_name = list(uploaded.keys())[0]
salary = pd.read_csv(file_name)

salary.head()
```

[Choose Files](#) Salary_Data...ords (1).csv

Salary_Data_500_Records (1).csv(text/csv) - 7015 bytes, last modified: 7/6/2026 - 100% done
 Saving Salary_Data_500_Records (1).csv to Salary_Data_500_Records (1) (2).csv

	Experience Years	Salary
0	7.5	97851.07
1	19.0	214014.05
2	14.6	165505.09
3	12.0	125077.16
4	3.1	45569.02

Next steps: [Generate code with salary](#) [New interactive sheet](#)

```
print(salary.shape)
print(salary.info())
print(salary.describe())
print(salary.isnull().sum())
```

```
(500, 2)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Experience Years  500 non-null    float64
```

```

1    Salary          500 non-null    float64
dtypes: float64(2)
memory usage: 7.9 KB
None
      Experience Years      Salary
count      500.000000      500.000000
mean         9.970600    115052.649740
std          5.972475     52603.823545
min          0.100000     25000.000000
25%          4.800000     69901.620000
50%         10.250000    116395.555000
75%         15.125000    159862.995000
max         19.900000    219610.490000
Experience Years      0
Salary                0
dtype: int64

```

```

X = salary[['Experience Years']]
y = salary['Salary']

```

```

print(X.head())
print(y.head())

```

```

      Experience Years
0              7.5
1             19.0
2             14.6
3             12.0
4              3.1
0      97851.07
1     214014.05
2     165505.09
3     125077.16
4      45569.02
Name: Salary, dtype: float64

```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

```

```

print(X_train.shape)
print(X_test.shape)

```

```

(400, 1)
(100, 1)

```

```

model = LinearRegression()

model.fit(X_train, y_train)

print("Intercept:", model.intercept_)
print("Coefficient:", model.coef_[0])

```

Intercept: 29383.45723077451
Coefficient: 8577.927225687961

```
y_pred = model.predict(X_test)

print("MAE :", mean_absolute_error(y_test, y_pred))
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("R2 Score:", r2_score(y_test, y_pred))
```

MAE : 9693.811268433077
MSE : 146812459.69404542
RMSE: 12116.619152801883
R2 Score: 0.9452551653797014

```
plt.figure(figsize=(8,5))

plt.scatter(X_test, y_test, color='blue', label='Actual Sala
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Re

plt.xlabel("Experience Years")
plt.ylabel("Salary")
plt.title("Simple Linear Regression")
plt.legend()

plt.show()
```



